Framework for Combining Randomized and Observational Data under Unmeasured Confounding

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Executive summary

We develop a Bayesian nonparametric framework for estimating heterogeneous treatment effects by combining data from randomized controlled trials and observational studies. The approach does not require unconfoundedness of the observational data; instead, unmeasured confounding is explicitly modeled through a confounding function. The outcome is decomposed into a prognostic baseline, a causal treatment effect, and a confounding bias, each modeled by Bayesian Additive Regression Trees. To accommodate block-wise and sporadic missing covariates arising from data fusion, we propose *informed random splitting*, a novel method for handling missing data within BART that makes stochastic routing decisions at ambiguous tree nodes using posterior predictive matching. We show that informed random splitting corresponds to exact Gibbs sampling under a well-defined augmented posterior, thereby providing valid uncertainty quantification without requiring an external imputation model.

1 Setup and notation

Consider a setting in which data are available from two sources: a randomized controlled trial (RCT) and an observational study (OS). Let $S \in \{0,1\}$ denote the data-source indicator, with $S = 1$ for the RCT and $S = 0$ for the OS. Let $A \in \{0,1\}$ denote the binary treatment assignment, $X \in \mathcal{X} \subseteq \mathbb{R}^p$ a vector of observed baseline covariates, and $U \in \mathcal{U}$ a vector of unmeasured covariates. Write $T(a)$ for the potential survival time under treatment a, and define $Y(a) = \log T(a)$.

2 Accelerated failure time model

We posit the following structural model for the potential outcomes:

$$Y(a) \;=\; f_0(X,U) \;+\; a\,\tau^*(X,U) \;+\; \sigma\,\varepsilon, \qquad \varepsilon \perp (X,U,A,S), \tag{1}$$

where $f_0(X,U)$ is the baseline (prognostic) function under control, $\tau^*(X,U) = \mathbb{E}[Y(1) - Y(0) \mid X,U]$ is the unit-level treatment effect, $\sigma > 0$ is a scale parameter, and ε has a mean-zero distribution with unit variance.

The observed outcome is $Y = Y(A)$ under the consistency assumption below.

3 Assumptions

We impose the following conditions. Assumptions 1–3 concern the internal validity of the RCT. Assumption 4 links the two data sources.

Assumption 1 (Consistency / SUTVA)**.** *The observed outcome satisfies* $Y = A\,Y(1) + (1 - A)\,Y(0)$*, and there is no interference between units.*

Assumption 2 (Randomization in the RCT)**.** *Within the RCT, treatment is randomized and hence independent of all pre-treatment variables:*

$$A \;\perp\; \big(Y(0),\, Y(1),\, U\big) \;\Big|\; X,\; S = 1. \tag{2}$$

Assumption 3 (Positivity of the RCT)**.** *For all* x *in the support of* $X \mid S = 1$ *and* $a \in \{0, 1\}$*,*

$$0 \;<\; \mathbb{P}(A = a \mid X = x,\, S = 1) \;<\; 1. \tag{3}$$

Assumption 4 (Transportability of potential outcomes)**.** *The conditional distribution of potential outcomes given observed covariates is the same across data sources:*

$$Y(a) \mid X,\, S = 1 \;\overset{d}{=}\; Y(a) \mid X,\, S = 0, \qquad a \in \{0, 1\}. \tag{4}$$

Remark 1 (No unconfoundedness in the OS)*.* We deliberately *do not* assume unconfoundedness in the observational study. That is, we allow

$$A \not\perp \big(Y(0),\, Y(1)\big) \;\Big|\; X,\; S = 0 \tag{5}$$

in general. This arises because treatment assignment in the OS may depend on the unmeasured covariates U:

$$\mathbb{P}(A = 1 \mid X, U, S = 0) \;\neq\; \mathbb{P}(A = 1 \mid X, S = 0) \quad \text{in general.} \tag{6}$$

Remark 2 (Covariate overlap)*.* For the data fusion to be informative, we require sufficient overlap between the covariate distributions in the two sources, i.e. the support of $X \mid S = 1$ should cover the support of $X \mid S = 0$ in regions where treatment effect estimation is desired.

4 Causal estimands

Definition 1 (Conditional average treatment effect)**.** The conditional average treatment effect (CATE) is

$$\tau(x) \;:=\; \mathbb{E}\big[Y(1) - Y(0) \mid X = x\big]. \tag{7}$$

Under the structural model (1), the CATE can be written as

$$\tau(x) \;=\; \mathbb{E}\big[\tau^*(X, U) \mid X = x\big], \tag{8}$$

which marginalizes the unit-level effect $\tau^*(X, U)$ over the conditional distribution of $U \mid X$.

By Assumption 4, $\tau(x)$ does not depend on the data source:

$$\mathbb{E}\big[Y(1) - Y(0) \mid X = x,\, S = 1\big] \;=\; \mathbb{E}\big[Y(1) - Y(0) \mid X = x,\, S = 0\big] \;=\; \tau(x). \tag{9}$$

In the survival setting, causal contrasts are expressed through the potential survival functions $S_a(t \mid x) = \mathbb{P}(T(a) > t \mid X = x)$. We consider:

$$\Delta_{\mathrm{RD}}(t, x) \;=\; S_1(t \mid x) - S_0(t \mid x), \tag{10}$$

$$\Delta_{\mathrm{RMST}}(t^*, x) \;=\; \int_0^{t^*} \big\{S_1(u \mid x) - S_0(u \mid x)\big\}\, \mathrm{d}u, \tag{11}$$

representing the risk difference at time t and the difference in restricted mean survival time up to t^*, respectively.

5 The confounding function

Because unconfoundedness does not hold in the OS, the observed treatment–control contrast in the OS does not equal the CATE. The discrepancy is captured by the following quantity.

Definition 2 (Confounding function)**.** The confounding function is

$$c(x) \;:=\; \mathbb{E}[Y \mid X = x,\, A = 1,\, S = 0] \;-\; \mathbb{E}[Y \mid X = x,\, A = 0,\, S = 0] \;-\; \tau(x). \tag{12}$$

The confounding function admits an explicit representation in terms of the unmeasured covariates.

Proposition 1 (Structural form of the confounding function)**.** *Under the structural model* (1) *and Assumptions 1–4, the confounding function satisfies*

$$c(x) \;=\; \mathbb{E}\big[f_0(X, U) \mid X = x,\, A = 1,\, S = 0\big] \;-\; \mathbb{E}\big[f_0(X, U) \mid X = x,\, A = 0,\, S = 0\big] \;+\; \delta_\tau(x), \tag{13}$$

where

$$\delta_\tau(x) \;=\; \mathbb{E}\big[\tau^*(X, U) \mid X = x,\, A = 1,\, S = 0\big] \;-\; \tau(x). \tag{14}$$

If the unit-level treatment effect does not depend on U, i.e. $\tau^*(X, U) = \tau(X)$, then $\delta_\tau(x) = 0$ and the confounding function reduces to the imbalance in the baseline prognostic function:

$$c(x) \;=\; \mathbb{E}\big[f_0(X, U) \mid X = x,\, A = 1,\, S = 0\big] \;-\; \mathbb{E}\big[f_0(X, U) \mid X = x,\, A = 0,\, S = 0\big]. \tag{15}$$

Proof. By Assumption 1 and the structural model (1),

$$\mathbb{E}[Y \mid X{=}x, A{=}1, S{=}0] \;=\; \mathbb{E}\big[f_0(X, U) + \tau^*(X, U) \mid X{=}x, A{=}1, S{=}0\big], \tag{16}$$

$$\mathbb{E}[Y \mid X{=}x, A{=}0, S{=}0] \;=\; \mathbb{E}\big[f_0(X, U) \mid X{=}x, A{=}0, S{=}0\big]. \tag{17}$$

Subtracting (17) from (16) yields

$$\begin{aligned}&\mathbb{E}[Y \mid X{=}x, A{=}1, S{=}0] - \mathbb{E}[Y \mid X{=}x, A{=}0, S{=}0] \\ &\qquad = \mathbb{E}\big[f_0(X,U) \mid X{=}x, A{=}1, S{=}0\big] - \mathbb{E}\big[f_0(X,U) \mid X{=}x, A{=}0, S{=}0\big] \\ &\qquad\qquad + \mathbb{E}[\tau^*(X,U) \mid X{=}x, A{=}1, S{=}0].\end{aligned}$$

By definition (12), subtracting $\tau(x) = \mathbb{E}[\tau^*(X,U) \mid X = x]$ gives

$$c(x) = \mathbb{E}\big[f_0(X,U) \mid X{=}x, A{=}1, S{=}0\big] - \mathbb{E}\big[f_0(X,U) \mid X{=}x, A{=}0, S{=}0\big] + \delta_\tau(x),$$

as claimed. When $\tau^*(X,U) = \tau(X)$, the conditional expectation given (X,A,S) equals $\tau(x)$, so $\delta_\tau(x) = 0$. □

Remark 3 (Interpretation). Equation (13) decomposes the confounding function into two sources of bias: (i) an imbalance in the baseline prognostic function $f_0(X,U)$ across treatment arms in the OS, arising because treated and control subjects have systematically different distributions of U given X; and (ii) a term $\delta_\tau(x)$ that captures effect modification by U, which is nonzero when the treatment effect itself depends on unobservables. Under the simplifying condition $\tau^*(X,U) = \tau(X)$—i.e. no unobserved effect modification—the confounding function reflects purely baseline prognostic imbalance.

6 Outcome decomposition

We now derive the key decomposition that motivates our modeling strategy.

Proposition 2 (Decomposition of the conditional outcome)**.** *Under Assumptions 1–4, the conditional expectation of the observed outcome admits the decomposition*

$$\mathbb{E}[Y \mid A, X, S] \;=\; m_0(X,S) \;+\; \tau(X)\,A \;+\; (1-S)\,A\,c(X), \tag{18}$$

where $m_0(X,S) := \mathbb{E}[Y \mid A = 0, X, S]$ *is the baseline outcome model.*

Proof. Case $A = 0$*.* By definition, $\mathbb{E}[Y \mid A{=}0, X, S] = m_0(X,S)$, and the right-hand side of (18) reduces to $m_0(X,S)$.

Case $A = 1$*,* $S = 1$ *(RCT).* By Assumptions 1 and 2,

$$\begin{aligned}\mathbb{E}[Y \mid A{=}1, X, S{=}1] \;&=\; \mathbb{E}[Y(1) \mid X, S{=}1] \\ &=\; \mathbb{E}[Y(0) \mid X, S{=}1] + \mathbb{E}[Y(1) - Y(0) \mid X, S{=}1] \\ &=\; m_0(X,1) + \tau(X),\end{aligned} \tag{19}$$

where the last equality uses Assumption 2 (so that $\mathbb{E}[Y(0) \mid X, S{=}1] = \mathbb{E}[Y \mid A{=}0, X, S{=}1] = m_0(X,1)$) and Assumption 4 (so that the conditional treatment effect equals $\tau(x)$). The right-hand side of (18) with $A = 1$, $S = 1$ gives $m_0(X,1) + \tau(X)$, matching (19).

Case $A = 1$*,* $S = 0$ *(OS).* By the definition of $c(X)$ in (12),

$$\mathbb{E}[Y \mid A{=}1, X, S{=}0] \;=\; m_0(X,0) + \tau(X) + c(X),$$

which matches the right-hand side of (18) with $A = 1$, $S = 0$. $\square$

The individual-level regression model follows directly:

$$Y_i \;=\; m_0(X_i, S_i) \;+\; A_i\,\tau(X_i) \;+\; A_i\,(1 - S_i)\,c(X_i) \;+\; \sigma\,\varepsilon_i. \tag{20}$$

7 Identification

Proposition 3 (Identification of τ and c)**.** *Under Assumptions 1–4:*

(i) *The CATE $\tau(x)$ is identified from the RCT data alone:*

$$\tau(x) \;=\; \mathbb{E}[Y \mid X{=}x, A{=}1, S{=}1] - \mathbb{E}[Y \mid X{=}x, A{=}0, S{=}1]. \tag{21}$$

(ii) *Given identification of $\tau(x)$, the confounding function $c(x)$ is identified from the OS data:*

$$c(x) \;=\; \mathbb{E}[Y \mid X{=}x, A{=}1, S{=}0] - \mathbb{E}[Y \mid X{=}x, A{=}0, S{=}0] - \tau(x). \tag{22}$$

(iii) *Neither $\tau(x)$ nor $c(x)$ is identified from the OS data alone.*

Proof. Part (i) follows from Assumptions 2 and 4, as shown in (19). Part (ii) is immediate from Definition 2. For part (iii), note that the OS data identify only the composite $\tau(x) + c(x)$; without external information (the RCT), the two components are not separately identifiable. $\square$

Remark 4 (Role of the observational data). Although $\tau(x)$ is identifiable from the RCT alone, joint estimation using both sources can improve efficiency. The decomposition (20) enables the OS to contribute to the estimation of $\tau(x)$ through the shared baseline function $m_0(X, S)$ and through the additional treated observations, while the confounding function $c(x)$ absorbs the bias. The efficiency gain is largest when $c(x)$ is small or smoothly varying, so that few degrees of freedom are consumed by the bias correction.

8 Accelerated failure time formulation and survival estimands

Substituting $Y_i = \log T_i$ and denoting by Φ the CDF of the (nonparametric) error distribution, the potential survival functions take the form

$$S_1(t \mid x, s) \;=\; 1 - \Phi\left(\frac{\log t - m_0(x, s) - \tau(x) - (1 - s)\,c(x)}{\sigma}\right), \tag{23}$$

$$S_0(t \mid x, s) \;=\; 1 - \Phi\left(\frac{\log t - m_0(x, s)}{\sigma}\right). \tag{24}$$

When the target population is the OS ($s = 0$), the survival-scale estimands become

$$\Delta_{\mathrm{RD}}(t, x) \;=\; \Phi\left(\frac{\log t - m_0(x, 0)}{\sigma}\right) - \Phi\left(\frac{\log t - m_0(x, 0) - \tau(x) - c(x)}{\sigma}\right), \tag{25}$$

$$\Delta_{\mathrm{RMST}}(t^*, x) \;=\; \int_0^{t^*} \left\{\Phi\left(\frac{\log u - m_0(x, 0)}{\sigma}\right) - \Phi\left(\frac{\log u - m_0(x, 0) - \tau(x) - c(x)}{\sigma}\right)\right\} \mathrm{d}u. \tag{26}$$

Remark 5 (Causal estimands in the target population). In equations (25)–(26), the confounding function $c(x)$ appears because the target population is the OS ($s = 0$). If instead one targets the RCT population ($s = 1$), the $c(x)$ term drops out and the estimands depend only on $m_0(x, 1)$ and $\tau(x)$.

Remark 6 (Causal interpretation of the survival estimands). The risk difference (25) and RMST difference (26) involving $c(x)$ reflect the *causal* treatment effect in the OS population. The confounding function enters through the survival curve under treatment, S_1, because the observed outcomes of the treated group in the OS are distorted by unmeasured confounding. The decomposition (20) ensures that $\tau(x)$ captures the true causal effect while $c(x)$ absorbs the bias, so that the survival contrasts computed from (23)–(24) with c removed yield the correct causal comparison.